

Notes: Butler, Mayer & Weston (RFS, 2023)

Racial Disparities in the Auto Loan Market

Contents

1	Big picture	3
2	Data and measurement	4
2.1	Why matching credit bureau and HMDA data matters	4
2.2	Matched sample and population covered	4
2.3	Race and ethnicity	4
2.4	Approval, interest rates, and default	4
2.5	Controls	5
2.6	Racial bias, competition, and institutional channels	5
3	Models and empirical specifications	6
3.1	Match selection test	6
3.2	Auto credit approval	6
3.3	Cross-sectional heterogeneity	6
3.4	Credit card falsification test	6
3.5	Auto loan interest rates	7
3.6	Default test	7
3.7	CFPB intervention difference-in-differences	7
4	Identification and empirical design	8
4.1	Why the paper uses triangulation	8
4.2	Why high racial-bias and low-competition areas matter	8
4.3	Why credit card lending is a falsification test	8
4.4	Why default matters	8
4.5	Why the CFPB intervention matters	9
4.6	Main limitations	9
5	Tables and figures	10
5.1	Table 1: Summary of the credit bureau/HMDA match	10
5.2	Table 2: Does borrower race affect the match?	11
5.3	Table 3: Summary statistics for the matched panel	11
5.4	Table 4: Applicant race and auto credit approval	12
5.5	Table 5, Figure 1, and Figure 2: Where is discrimination strongest?	13
5.6	Table 6: Credit card falsification test	13
5.7	Table 7 and Table 8: Auto loan summary statistics and interest-rate disparities	14
5.8	Table 9: Borrower race and auto loan default rates	16
5.9	Table 10 and Figure 3: The 2013 CFPB intervention	17

6	Short summary	17
7	Conclusion	19
7.1	Core conclusion	19
7.2	Why the paper is persuasive	19
7.3	Economic intuition	19
7.4	Extensive versus intensive margin	19
7.5	Policy implications	19
7.6	Caveats and external validity	19
7.7	Takeaway	20

1 Big picture

- **Question.** Are there racial disparities in the auto loan market after controlling for borrower creditworthiness, geography, demographics, loan characteristics, and time effects? If yes, are the disparities best interpreted as discrimination rather than omitted risk or statistical discrimination?
- **Market setting.** Auto lending is a large consumer credit market, but it is less transparent than mortgage lending. Lenders do not report race for auto loans, and dealer-intermediated lending creates scope for discretion in approval and pricing.
- **Data innovation.** The paper links a 1% nationally representative credit bureau panel to Home Mortgage Disclosure Act (HMDA) mortgage records. The credit bureau provides auto credit outcomes and financial controls; HMDA provides race and ethnicity.
- **Extensive-margin result.** Black and Hispanic applicants are less likely to obtain auto credit. With full controls, the minority approval gap is about 1.5 percentage points, and the gap is larger for subprime applicants.
- **Aggregate magnitude.** A back-of-the-envelope calculation implies that more than 80,000 minority borrowers each year fail to obtain loans they would have obtained if they were white, holding observables fixed.
- **Cross-sectional evidence.** Approval disparities are stronger in high racial-bias states and in markets with lower lending competition, patterns consistent with preference-based or stereotype-based discrimination.
- **Falsification test.** The same borrowers do not show similar racial disadvantages in credit card lending, a more automated market with less face-to-face discretion.
- **Intensive-margin result.** Minority borrowers who do get auto loans pay about 70 basis points more in interest rates after controls. The gap is larger in high racial-bias states.
- **Default result.** Conditional on observables, minorities default no more, and subprime minority borrowers default less. This is difficult to reconcile with omitted risk or accurate statistical discrimination.
- **Regulatory result.** The 2013 CFPB intervention reduced unexplained racial differences in auto loan rates by roughly 60%, without a clear reduction in minority credit access.
- **Main economic takeaway.** The paper triangulates discrimination using approval gaps, interest-rate gaps, geography, competition, credit card falsification, default outcomes, and regulatory timing. No single test is definitive, but the combined evidence strongly supports discriminatory treatment in auto lending.

2 Data and measurement

2.1 Why matching credit bureau and HMDA data matters

- Auto lenders are not required to report borrower race in the same way mortgage lenders report it under HMDA.
- The authors therefore infer race by linking credit bureau records to HMDA mortgage records.
- Mortgages appear in both data sets with enough detail to match many credit bureau mortgages to HMDA applications.
- Once a person is matched, the authors can observe race from HMDA and auto credit outcomes from the credit bureau panel.

Interpretation: The paper’s key measurement challenge is not estimating regressions; it is constructing a data set that contains both race and auto lending outcomes. Table 1 and Table 2 exist to show that this matching procedure is feasible and not obviously racially biased.

2.2 Matched sample and population covered

- The matched sample is drawn from mortgage borrowers, so it is not the full population of auto loan applicants.
- The sample is likely more creditworthy than the general auto-loan-applicant population.
- This selection likely makes the strongest discrimination effects harder to observe, because the paper finds the largest effects among subprime borrowers and the matched sample has relatively better credit.

Interpretation: The results should be read as evidence for homeowners or recent mortgage borrowers with observable auto credit activity. The authors argue that the effects may understate the broader auto-market disparities because very risky borrowers are underrepresented.

2.3 Race and ethnicity

- The paper uses “race” to refer to both race and ethnicity.
- The sample is limited to white, Black, and Hispanic borrowers.
- Black and Hispanic borrowers are grouped into a minority indicator for most tests.

Interpretation: Combining Black and Hispanic borrowers increases power and creates a single main effect, while robustness checks and interactions help show whether results differ by subgroup.

2.4 Approval, interest rates, and default

- Auto credit approval is measured using credit bureau information on whether an applicant successfully opens a new auto loan.
- The approval measure combines lender approval and borrower take-up. This is important because the credit bureau cannot observe rejected applications in the same direct way a lender can.
- Auto loan interest rates are observed on originated loans.
- Default is observed after origination and is used to test whether minority borrowers are actually riskier conditional on controls.

Interpretation: Approval and pricing examine discrimination before and at loan origination. Default examines whether the loans to minority borrowers were worse risks ex post.

2.5 Controls

The main regressions use a rich set of controls:

- demographics: age, sex, income;
- credit characteristics: lagged credit score, total debt, debt-to-income, past due debt, auto debt share;
- ZIP-code characteristics: local income, density, commuting by car, banking access, and other local measures;
- fixed effects: state-by-year fixed effects, origination month indicators, and indicators for time relative to the credit bureau/HMDA match;
- loan characteristics in rate and default tests: loan amount, loan term, auto loan-to-income ratio, and related loan terms.

Interpretation: The richness of controls is central. A racial coefficient after these controls is not a raw gap; it is a residual gap among applicants or borrowers who are similar on observable credit, income, geography, and time.

2.6 Racial bias, competition, and institutional channels

- Racial bias is measured using Google searches for racial slurs, following Stephens-Davidowitz (2014).
- Banking competition and dealer competition proxy for the competitive pressure that may discipline discriminatory behavior.
- Nonbank financing share helps identify where the CFPB intervention should matter most, because the 2013 guidance focused on indirect auto lending.

Interpretation: These variables help distinguish discrimination from omitted credit risk. If the racial gaps are larger where racial animus is higher and competition is lower, the patterns are harder to explain as simple unobserved risk differences.

3 Models and empirical specifications

3.1 Match selection test

The matching-selection regressions can be summarized as:

$$\Pr(\text{Matched}_i = 1) = \alpha + \beta_1 \text{Black}_i + \beta_2 \text{Hispanic}_i + \Gamma X_i + \delta_{tract \times year} + \varepsilon_i. \quad (1)$$

Interpretation: The goal is not to estimate discrimination here. The goal is to check whether the construction of the matched sample creates strong race-related selection. Table 2 shows little evidence that Black or Hispanic status systematically drives match success once other mortgage characteristics are controlled.

3.2 Auto credit approval

The baseline approval equation is:

$$\text{Approval}_{it} = \alpha + \beta \text{Minority}_i + \Gamma X_{i,t-1} + \Lambda Z_{zip,t} + \mu_{state \times year} + \tau_{matchtime} + u_{it}. \quad (2)$$

- Approval_{it} equals one if the applicant successfully opens a new auto loan.
- Minority_i equals one for Black or Hispanic borrowers.
- $X_{i,t-1}$ includes individual credit and demographic controls.
- $Z_{zip,t}$ includes local ZIP-code controls.

Interpretation: The coefficient β is the residual approval gap between otherwise similar minority and white applicants. Because approval includes take-up, the coefficient should be interpreted as a racial gap in successful credit access, not as a pure lender-approval-only estimate.

3.3 Cross-sectional heterogeneity

The paper tests whether the minority gap is larger where discrimination is more likely:

$$\text{Approval}_{it} = \alpha + \beta \text{Minority}_i + \theta(\text{Minority}_i \times H_g) + \Gamma X_{it} + \Lambda Z_{zip,t} + FE + u_{it}, \quad (3)$$

where H_g is a market characteristic such as high racial bias, low banking competition, low dealer competition, high nonbank financing, or rural location.

Interpretation: If the unexplained racial gap reflects discrimination, θ should be negative in high-bias or low-competition environments. This is what the paper finds for racial bias and competition.

3.4 Credit card falsification test

For the same borrowers, the paper estimates:

$$\text{CCLimitIncrease}_{it} = \alpha + \beta \text{Minority}_i + \theta(\text{Minority}_i \times H_g) + \Gamma X_{it} + \Lambda Z_{zip,t} + FE + u_{it}. \quad (4)$$

Interpretation: Credit card credit-limit increases are more automated and less dealer-mediated than auto lending. If the auto results were just omitted creditworthiness, similar racial patterns should appear in credit cards. Table 6 shows that they do not.

3.5 Auto loan interest rates

For originated loans, the interest-rate equation is:

$$\text{Rate}_{i\ell t} = \alpha + \beta \text{Minority}_i + \Gamma X_{i,t-1} + \Pi L_{\ell t} + \Lambda Z_{zip,t} + \mu_{state \times year} + \kappa_{month} + \tau_{matchtime} + u_{i\ell t}, \quad (5)$$

where $L_{\ell t}$ includes loan characteristics such as amount, term, and auto-loan-to-income ratio.

Interpretation: This regression asks whether minority borrowers pay higher rates even among borrowers who were actually approved and after controlling for loan terms and risk characteristics.

3.6 Default test

The ex post default regression is:

$$\text{Default}_{i\ell t} = \alpha + \beta \text{Minority}_i + \Gamma X_{i,t-1} + \Pi L_{\ell t} + \Lambda Z_{zip,t} + FE + u_{i\ell t}. \quad (6)$$

Interpretation: If minority borrowers paid higher rates because they were unobservably riskier, they should default more after controls. Instead, the paper finds lower default rates for subprime minority borrowers.

3.7 CFPB intervention difference-in-differences

The 2013 regulatory intervention is studied with:

$$Y_{it} = \alpha + \beta_1 \text{Minority}_i + \beta_2 (\text{Minority}_i \times \text{Post}_t) + \Gamma X_{it} + FE + u_{it}, \quad (7)$$

where Y is auto loan rate, auto credit approval, or credit card limit increase.

A triple-difference version tests whether the effect is stronger in high nonbank financing areas:

$$Y_{it} = \alpha + \beta (\text{Minority}_i \times \text{Post}_t \times \text{HighNonbank}_g) + \text{lower-order terms} + \Gamma X_{it} + FE + u_{it}. \quad (8)$$

Interpretation: The CFPB guidance targeted indirect auto lending, so a decline in interest-rate disparities after 2013, especially in high nonbank areas, supports the interpretation that discretionary pricing was being constrained by regulation.

4 Identification and empirical design

4.1 Why the paper uses triangulation

- Racial discrimination cannot be observed directly in the data.
- A residual racial coefficient alone could reflect omitted variables, imperfect risk controls, or true discrimination.
- The paper therefore combines several pieces of evidence: approval gaps, pricing gaps, racial bias measures, competition measures, a credit card falsification test, ex post default outcomes, and a regulatory shock.

Interpretation: The empirical strategy is cumulative. Each piece has limits, but the set of patterns is difficult to reconcile with a single omitted creditworthiness variable.

4.2 Why high racial-bias and low-competition areas matter

- Preference-based discrimination should be stronger where racial animus is more prevalent.
- Competitive pressure should reduce discriminatory behavior if lenders or dealers give up profitable minority loans because of bias.
- Statistical discrimination based on accurate beliefs should be less sensitive to competition, because it is profit-maximizing.

Interpretation: The finding that racial gaps are larger in high-bias and low-competition markets is one of the paper's strongest discriminating tests between bias and accurate statistical discrimination.

4.3 Why credit card lending is a falsification test

- The same borrowers appear in both auto credit and credit card data.
- Credit card credit-limit increases are more automated and less dependent on face-to-face interaction with dealers or loan officers.
- If unobserved credit risk explains auto lending disparities, credit card lenders should also disadvantage the same minority borrowers.

Interpretation: The absence of parallel credit-card disparities weakens omitted-risk explanations and points to discretionary features of auto lending.

4.4 Why default matters

- If minority borrowers are unobservably riskier, they should default more after approval, especially when charged higher rates.
- The paper finds no such pattern; subprime minority borrowers default less conditional on observables.
- Lower default among marginal minority borrowers implies that lenders may be rejecting or overpricing borrowers who are actually profitable.

Interpretation: The default evidence directly challenges the claim that higher rates or lower approvals simply price unobserved risk.

4.5 Why the CFPB intervention matters

- The CFPB increased scrutiny of indirect auto lending in 2013.
- If higher minority rates are discretionary, regulatory scrutiny should reduce the unexplained rate gap.
- If higher rates are needed to compensate for risk, lowering them should reduce access or worsen lender outcomes.
- The paper finds the rate gap fell substantially without an accompanying credit-access contraction.

Interpretation: The intervention provides a policy-based test: disparities respond to enforcement pressure, which is consistent with discrimination rather than unavoidable risk pricing.

4.6 Main limitations

- Race is observed only for people who can be matched to HMDA mortgage records.
- Approval is measured as successful loan origination, so it combines lender approval and borrower take-up.
- The data do not observe every detail of dealer bargaining, vehicle characteristics, lender underwriting rules, or borrower search behavior.
- The paper cannot run a randomized audit study, so it relies on observational controls and multiple falsification/validation tests.

Interpretation: These limitations do not overturn the evidence, but they explain why the paper emphasizes triangulation rather than claiming that any one regression proves discrimination by itself.

5 Tables and figures

5.1 Table 1: Summary of the credit bureau/HMDA match

Read:

- Home purchase mortgages have a 61.96% match rate; refinance loans have an 80.12% match rate; the total match rate is 68.82%.
- Matched and unmatched borrowers are not identical. For example, matched purchase borrowers are slightly younger and have somewhat lower total debt than unmatched borrowers.
- Normalized differences are generally modest for many mortgage characteristics, but some credit-bureau variables show meaningful selection, especially whether the borrower already had a mortgage.

Interpretation: The table establishes the data construction. The paper does not simply assume that credit bureau and HMDA records can be linked; it documents how often the link succeeds and how matched borrowers compare to unmatched borrowers. The main empirical sample is therefore not a random draw of all consumers. It is a selected, matched panel of mortgage-linked borrowers whose race can be observed. This matters because the sample likely has better credit and more formal financial-market experience than the typical auto borrower.

Economic message: The table supports the feasibility of the paper’s data innovation while reminding the reader that external validity is limited. If discrimination is strongest for riskier borrowers, the matched homeowner sample may understate the magnitude of disparities in the broader auto market.

What not to over-read: Do not read Table 1 as proving that the matched sample is perfectly representative. It shows that the match is large and usable, not that selection is zero.

Table 1
Summary of the credit bureau/HMDA match

A. Match rate

	Credit bureau sample	Matched to HMDA	Match rate
Home purchase mortgages	107,085	66,345	61.96%
Refinance loans	65,046	52,115	80.12%
All loans	172,131	118,460	68.82%

B. Home purchase mortgages

	Credit bureau sample (N = 107,085)	Matched to HMDA (N = 66,345)	Unmatched (N = 40,740)	Matched vs. Unmatched	
				Norm. diff.	t-stat
<i>Match criteria</i>					
Conventional loan	0.631	0.623	0.643	-0.03	-4.70
FHA loan	0.289	0.293	0.283	0.01	2.47
VA loan	0.080	0.084	0.074	0.03	5.38
Fannie Mae	0.243	0.251	0.231	0.03	5.89
Freddie Mac	0.149	0.158	0.134	0.05	9.98
Loan amount	192,142	193,758	189,508	0.02	3.99
<i>Nonmatch characteristics</i>					
Credit score _{t-1}	717	719	715	0.04	7.78
Age	42.0	41.1	43.3	-0.12	-21.97
Have mortgage _{t-1}	0.310	0.254	0.401	-0.23	-33.37
Total debt _{t-1}	78,802	66,519	98,895	-0.19	-24.01
Past due debt _{t-1}	311	283	356	-0.02	-3.43
Auto debt _{t-1}	8,176	8,145	8,227	-0.00	-1.00

C. Refinance loans

	Credit bureau sample (N = 65,046)	Matched to HMDA (N = 52,115)	Unmatched (N = 12,931)	Matched vs. Unmatched	
				Norm. diff.	t-stat
<i>Match criteria</i>					
Conventional loan	0.815	0.814	0.821	-0.01	-1.85
FHA loan	0.125	0.125	0.124	0.00	0.27
VA loan	0.060	0.061	0.055	0.02	2.74
Fannie Mae	0.307	0.308	0.301	0.01	1.49
Freddie Mac	0.202	0.210	0.171	0.07	9.86
Loan amount	196,062	193,971	204,491	-0.06	-7.21
<i>Nonmatch characteristics</i>					
Credit score _{t-1}	738	738	739	-0.01	-1.04
Age	49.4	49.6	48.7	0.05	7.61
Have mortgage _{t-1}	1.00	1.00	1.00	.	.
Total debt _{t-1}	214,145	212,926	219,054	-0.03	-3.97
Past due debt _{t-1}	233	229	250	-0.00	-0.62
Auto debt _{t-1}	8,128	8,058	8,409	-0.02	-2.67

This table summarizes the match between mortgages in the credit bureau data and mortgages in the HMDA data.

Table 2
Does borrower race affect the credit bureau/HMDA match?

	t(Matched)		
	Full sample (1)	Home purchase mortgages (2)	Refinance loans (3)
<i>Match criteria</i>			
FHA loan	0.008 (0.006)	-0.116*** (0.008)	0.005 (0.010)
VA loan	0.057*** (0.009)	-0.025** (0.012)	0.021 (0.016)
Purchased by Fannie Mae	0.107*** (0.005)	0.212*** (0.010)	0.093*** (0.006)
Purchased by Freddie Mac	0.130*** (0.006)	0.281*** (0.013)	0.114*** (0.007)
log(Loan amount)	0.026*** (0.005)	-0.016* (0.009)	-0.003 (0.006)
<i>Nonmatch characteristics</i>			
Black	-0.157 (0.154)	-0.167 (0.225)	-0.345 (0.215)
Hispanic	-0.013 (0.129)	-0.320* (0.184)	0.137 (0.188)
Black X log(Income)	0.012 (0.014)	0.012 (0.020)	0.031 (0.019)
Hispanic X log(Income)	0.001 (0.012)	0.027 (0.017)	-0.011 (0.017)
log(Income)	-0.137*** (0.004)	-0.189*** (0.008)	-0.060*** (0.005)
Census tract-by-year FE	Yes	Yes	Yes
R-squared	.022	.044	.042
Observations	18,085,605	8,921,824	9,141,794

5.2 Table 2: Does borrower race affect the match?

Read:

- Black and Hispanic indicators are not strongly related to match success once mortgage characteristics, income, and census tract-by-year fixed effects are included.
- The Hispanic coefficient is marginally negative in the home-purchase column, but the overall pattern does not indicate a large systematic race-based matching problem.
- Mortgage characteristics such as Fannie Mae or Freddie Mac purchase status matter much more for match success than race.

Interpretation: This table addresses a crucial sample-selection concern: if minorities were systematically harder to match to credit bureau records, the resulting panel could bias later tests. The evidence suggests the matching process is not primarily sorting borrowers by race after controlling for observable mortgage information and geography.

Economic message: The credibility of the later approval and rate tests depends on race not being mechanically embedded in the match process. Table 2 reduces that concern.

What not to over-read: The table does not prove that sample selection is irrelevant. It only shows that race itself does not strongly predict match success in this matching specification.

5.3 Table 3: Summary statistics for the matched panel

Read:

- The matched sample is more financially established than the full credit bureau population: matched borrowers have higher credit scores and more mortgage exposure.
- Within the matched sample, white borrowers have higher credit scores and incomes than Black and Hispanic borrowers.
- Auto loan applicants show large raw differences in approval: white applicants have higher approval rates than Black or Hispanic applicants.
- The table also shows large racial differences in past due debt, auto debt, and debt-to-income measures.

Interpretation: Table 3 is the raw descriptive foundation for the paper. It shows that minorities and white borrowers differ on many observable dimensions. That is why the later regressions require rich controls. The table also illustrates why a raw approval gap alone is not enough: credit scores, income, and debt positions differ by race before regression adjustment.

Economic message: The descriptive facts justify both sides of the paper’s empirical design: (i) disparities are large enough to study, and (ii) the authors must control carefully for creditworthiness before interpreting residual gaps as possible discrimination.

What not to over-read: Do not interpret the raw racial differences as causal discrimination. Table 3 is descriptive and intentionally precedes the controlled regressions.

Table 3
Summary statistics for the Credit Bureau/HMDA matched panel
A. 2010 snapshot

	Full credit bureau sample (N=2,597,877)	Matched sample (N=78,932)	White (N=65,207)	Black (N=6,338)	Hispanic (N=7,387)
Credit score _{t-1}	669 [113]	707 [87.2]	715 [84.0]	660 [94.9]	678 [89.9]
Age	49.8 [18.9]	42.3 [13.9]	42.6 [14.1]	42.8 [13.5]	39.9 [12.9]
Have mortgage _{t-1}	0.295 [0.456]	0.552 [0.497]	0.577 [0.494]	0.431 [0.495]	0.428 [0.495]
Total debt _{t-1}	67,475 [164,108]	123,552 [166,047]	129,415 [170,688]	92,478 [125,459]	98,034 [148,536]
Past due debt _{t-1}	1,890 [12.611]	805 [4.750]	654 [4.319]	1,609 [6.797]	1,457 [5.991]
Auto debt _{t-1}	3,665 [8.917]	6,587 [11.019]	6,468 [10.958]	7,161 [11.065]	7,152 [11.478]
Income	.	73,295 [83,244]	75,805 [88,953]	62,686 [37,173]	60,239 [51,847]
Debt to income _{t-1}	.	1.86 [2.64]	1.89 [2.42]	1.54 [2.30]	1.82 [4.30]

B. Auto loan applicants (2005–2017)

	Full credit bureau sample (N=4,406,635)	Matched sample (N=218,476)	White (N=175,911)	Black (N=18,408)	Hispanic (N=24,157)
Credit approval (auto)	0.722 [0.448]	0.832 [0.374]	0.847 [0.360]	0.783 [0.412]	0.757 [0.429]
Credit score _{t-1}	663 [105]	697 [82.4]	705 [79.8]	655 [88.6]	673 [82.1]
Age	43.2 [14.9]	41.7 [13.1]	42.0 [13.2]	42.2 [12.9]	39.7 [12.3]
Have mortgage _{t-1}	0.401 [0.490]	0.643 [0.479]	0.661 [0.473]	0.560 [0.496]	0.569 [0.495]
Total debt _{t-1}	102,200 [193,180]	152,308 [185,190]	158,553 [192,014]	120,910 [132,993]	130,351 [162,920]
Past due debt _{t-1}	1,667 [8.360]	639 [4.725]	521 [4.663]	1,269 [4.779]	1,027 [5.066]
Auto debt _{t-1}	9,170 [15,190]	10,986 [15,752]	10,880 [15,748]	10,814 [15,159]	11,906 [16,191]
Income	.	78,395 [97,191]	81,578 [104,641]	65,480 [38,979]	65,061 [64,490]
Debt to income _{t-1}	.	2.18 [2.74]	2.18 [2.47]	1.98 [2.21]	2.31 [4.48]

This table presents summary statistics describing the Credit Bureau/HMDA matched panel (see Section 2.2 for

Table 4
Effect of applicant race on auto credit approval

	Credit approval (auto)				
	Full sample (1)	Subprime (2)	Prime (3)	Subprime (4)	Prime (5)
<i>Demographics</i>					
Minority	-4.465*** (0.289)	-1.480*** (0.259)	-1.661*** (0.332)	-2.375*** (0.399)	-0.840*** (0.271)
Minority X Hispanic			0.328 (0.410)		
Female	1.598*** (0.173)	1.115*** (0.169)	1.126*** (0.169)	1.492*** (0.352)	1.133*** (0.180)
Age	0.042*** (0.008)	-0.067*** (0.008)	-0.066*** (0.008)	0.017 (0.015)	-0.072*** (0.009)
log(Income)	3.886*** (0.184)	1.704*** (0.180)	1.711*** (0.181)	4.586*** (0.407)	0.736*** (0.199)
<i>Credit characteristics</i>					
Credit score _{t-1}		0.057*** (0.002)	0.057*** (0.002)	0.161*** (0.004)	0.013*** (0.003)
log(Total debt _{t-1})		0.866*** (0.053)	0.866*** (0.053)	0.403*** (0.070)	0.868*** (0.077)
Debt to income _{t-1}		-0.032 (0.062)	-0.032 (0.062)	0.040 (0.119)	-0.220*** (0.079)
log(Past due debt _{t-1})		-1.179*** (0.051)	-1.178*** (0.051)	-0.745*** (0.061)	-0.413*** (0.066)
<i>ZIP code characteristics</i>					
log(Personal income per capita)	1.087* (0.629)	-0.095 (0.611)	-0.076 (0.611)	0.573 (1.088)	-0.350 (0.701)
log(Population density)	-0.014 (0.065)	0.009 (0.065)	0.010 (0.065)	0.067 (0.142)	0.037 (0.072)
Bachelor's degree	5.108*** (1.254)	1.406 (1.236)	1.373 (1.238)	3.907 (2.374)	1.765 (1.372)
Commute using car	12.020*** (1.194)	10.569*** (1.149)	10.533*** (1.146)	12.663*** (2.317)	8.640*** (1.276)
State-by-year FE	Yes	Yes	Yes	Yes	Yes
Time relative to match indicators	Yes	Yes	Yes	Yes	Yes
R-squared	.047	.085	.085	.105	.047
Observations	218,300	214,534	214,534	68,494	146,036

5.4 Table 4: Applicant race and auto credit approval

Read:

- Without the full set of controls, the minority approval gap is about -4.465 percentage points.
- With demographics, credit controls, ZIP controls, state-year fixed effects, and match-time indicators, the main gap is about -1.480 percentage points.
- The Hispanic interaction is not significant, suggesting similar effects for Black and Hispanic applicants in this specification.
- The effect is much larger for subprime applicants: about -2.375 percentage points, versus about -0.840 percentage points for prime applicants.
- The authors interpret the -1.48 percentage point gap as comparable to a meaningful credit-score decline and estimate that it crowds out more than 80,000 minority loans annually.

Interpretation: Table 4 is the first main result. The coefficient remains negative even after the paper controls for a wide set of borrower and location characteristics. The larger gap among subprime borrowers is important because loan officer or dealer discretion is likely more influential when underwriting is less straightforward. This pattern is consistent with discriminatory discretion operating near the extensive margin of credit.

Economic message: Minority applicants are not merely different in raw observables. Conditional on observables, they are still less likely to obtain auto credit. The result quantifies the extensive-margin cost of discrimination: some minority borrowers who look similar to white borrowers do not get loans.

What not to over-read: The dependent variable is successful loan origination, not a direct lender approval decision. A lower success rate could include lender rejection, unfavorable offers that reduce take-up, or borrower response. The later tests help narrow the interpretation.

5.5 Table 5, Figure 1, and Figure 2: Where is discrimination strongest?

Read:

- Table 5 shows that the minority approval gap is much larger in high racial-bias states: the interaction with high racial bias is about -1.910 percentage points.
- The gap is also larger in lower competition markets: the interaction is negative for low banking competition and low dealer competition.
- The high nonbank financing and rural interactions are not the main drivers of approval disparities.
- Figure 1 plots state-level approval disparities against Google search volume for racial slurs and shows that states with higher measured bias tend to have larger minority approval reductions.
- Figure 2 maps where the evidence of reduced minority access is statistically strongest.

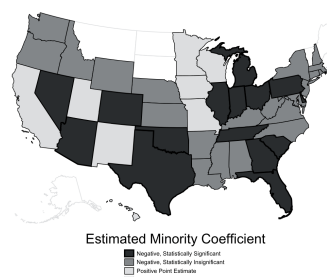
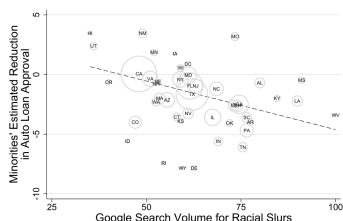
Interpretation: This group of evidence asks whether the unexplained approval gap behaves like discrimination. The racial-bias pattern is difficult to explain with a generic omitted variable unless that omitted variable also varies with racial animus in the same way. The competition result is also important: preference-based discrimination can survive more easily when lenders or dealers face less competitive pressure.

Economic message: The same residual approval gap is not uniform across places. It is stronger exactly where a discrimination theory predicts it should be stronger: places with more racial animus and less competitive discipline.

What not to over-read: Google search measures and competition proxies are not perfect measures of individual lender bias. They are market-level proxies. Their power comes from the cross-sectional pattern, not from measuring the attitudes of each dealer or lender.

Table 5
Where does applicant race have the largest impact on auto credit approval?

	Credit approval (auto)					
	(1)	(2)	(3)	(4)	(5)	(6)
Minority	-1.440*** (0.239)	-0.906*** (0.254)	-1.268*** (0.225)	-1.249*** (0.277)	-1.259*** (0.298)	-1.509*** (0.246)
Minority X High racial bias state			-1.910*** (0.443)			
Minority X Low banking competition			-0.728* (0.254)			
Low banking competition			0.214 (0.207)			
Minority X Low dealer competition			-0.783* (0.406)			
Low dealer competition			-0.118 (0.225)			
Minority X High nonbank financing			-0.351 (0.401)			
High nonbank financing			-0.782*** (0.461)			
Minority X Rural				0.117 (0.461)		
Rural				-0.124 (0.301)		
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
ZIP code controls	Yes	Yes	Yes	Yes	Yes	Yes
State-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Time relative to match indicators	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.085	.085	.085	.085	.085	.085
Observations	214,534	214,534	214,534	214,534	214,534	214,534



5.6 Table 6: Credit card falsification test

Read:

- The dependent variable is credit card limit increases for the same borrowers.
- Minority status does not show the same adverse pattern in credit card lending that appears in auto lending.
- The cross-sectional interactions with racial bias and competition are also absent or not economically comparable to the auto approval results.

Interpretation: Table 6 is a falsification test against an omitted-creditworthiness story. If minority borrowers were simply unobservably weaker credit risks, then credit card lenders should also treat them worse when evaluating credit-limit increases. The absence of parallel disparities suggests that the auto-lending channel is special, likely because it involves more human discretion and dealer intermediation.

Economic message: The table strengthens the paper’s discrimination interpretation by showing that the problem is not that the matched minority borrowers are universally less creditworthy across credit markets.

What not to over-read: Credit card lending and auto lending are not identical products. The falsification test does not prove that credit cards are discrimination-free; it shows that the specific auto-loan patterns do not simply replicate in another consumer credit market for the same people.

Table 6
Falsification test: Race and credit card lending

	CC limit increase					
	(1)	(2)	(3)	(4)	(5)	(6)
Minority	38.23 (73.09)	-10.44 (84.07)	110.36 (85.54)	54.49 (94.81)	94.97 (101.61)	7.12 (83.03)
Minority X High racial bias state		181.61 (154.99)				
Minority X Low banking competition			-234.78 (145.53)			
Low banking competition			74.16 (72.00)			
Minority X Low dealer competition				-62.18 (136.59)		
Low dealer competition				121.98 (79.09)		
Minority X High nonbank financing				-121.48 (135.60)		
High nonbank financing				179.29** (84.55)		
Minority X Rural					121.35 (155.43)	
Rural					50.05 (105.87)	
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
ZIP code controls	Yes	Yes	Yes	Yes	Yes	Yes
State-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Time relative to match indicators	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.075	.075	.075	.075	.075	.075
Observations	124,601	124,601	124,601	124,601	124,601	124,601

The tests in this table examine racial differences in credit card lending as a falsification test: we expect these lenders’ automated approach to reduce discrimination. Specifically, we select the subset of auto loan applicants who applied for credit cards (or credit card limit increases) during the same year as their auto loan application. We then regress the dollar increase in the borrower’s total credit card limit across all cards (average increase = \$3,090) on race, individual characteristics, and ZIP code characteristics. Following prior tests, the explanatory variables of interest are indicators for the applicant belonging to a racial minority, and the interaction of *Minority* with indicators for living in a state in the top tercile of racial bias (based on Google

Table 7
Summary statistics on the auto loans in the Credit Bureau/HMDA matched panel

	Full sample (N=25,697)	White borrowers (N=20,823)	Minority borrowers (N=4,874)	Subprime borrowers (N=6,115)	Prime borrowers (N=19,574)
<i>Demographics</i>					
Female	0.425 [0.494]	0.422 [0.494]	0.437 [0.496]	0.407 [0.491]	0.430 [0.495]
Age	43.5 [13.7]	43.7 [13.8]	42.7 [13.0]	40.2 [12.5]	44.5 [13.9]
Income	67,354 [40,075]	69,276 [41,296]	59,144 [33,143]	59,396 [32,920]	69,847 [41,758]
<i>Auto loan variables</i>					
Auto loan default	0.017 [0.130]	0.013 [0.114]	0.035 [0.184]	0.055 [0.228]	0.004 [0.066]
Auto loan rate	0.060 [0.048]	0.057 [0.045]	0.077 [0.058]	0.100 [0.063]	0.048 [0.034]
Auto loan amount	21,233 [10,201]	21,017 [10,178]	22,157 [10,244]	20,058 [9,897]	21,603 [10,266]
Auto-loan-to-income ratio	0.389 [0.248]	0.373 [0.238]	0.455 [0.279]	0.400 [0.248]	0.385 [0.249]
Auto loan term (months)	65.1 [13.2]	64.6 [13.2]	67.5 [12.7]	66.9 [13.6]	64.6 [13.0]
<i>Credit characteristics</i>					
Credit score _{t-1}	717 [78.4]	724 [75.3]	685 [83.3]	604 [44.7]	752 [47.0]
Total debt _{t-1}	129,567 [123,667]	133,584 [125,091]	112,407 [115,843]	96,101 [115,554]	140,053 [124,269]
Debt to income _{t-1}	2.08 [1.78]	2.09 [1.75]	2.06 [1.90]	1.73 [1.87]	2.20 [1.73]
Past due debt _{t-1}	308 [1,312]	237 [1,157]	609 [1,800]	1,129 [2,366]	51 [483]
Auto debt share	0.278 [0.311]	0.270 [0.306]	0.315 [0.329]	0.397 [0.372]	0.241 [0.279]

5.7 Table 7 and Table 8: Auto loan summary statistics and interest-rate disparities

Read:

- Table 7 describes originated auto loans. Minority borrowers have lower average credit scores and higher raw interest rates than white borrowers.
- Table 8 shows that after rich controls, minority borrowers pay about 0.704 percentage points more in auto loan rates in the main controlled specification.
- The raw gap is larger, about 1.600 percentage points, but much of that raw difference is explained by observables.
- The remaining controlled gap is still economically meaningful: roughly 70 basis points.
- The rate gap is larger in high racial-bias states and in low dealer competition areas.

Interpretation: Table 8 extends the analysis from access to pricing. Even after minority borrowers clear the approval hurdle, they pay higher rates. This is especially striking because Table 4 shows stricter approval on the extensive margin. If lenders approve fewer minority borrowers, the approved minority borrowers should be safer on average, all else equal. A persistent rate premium after approval therefore raises a serious concern about discretionary pricing.

Economic message: The discrimination cost is not only that some borrowers fail to obtain credit. Minority borrowers who do get loans also face higher borrowing costs, transferring surplus from borrowers to dealers/lenders and potentially reducing consumer welfare.

Table 8
Effect of borrower race on auto loan interest rates

	Auto loan rate						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Demographics and interaction terms</i>							
Minority	1.600*** (0.169)	0.704*** (0.117)	0.442*** (0.084)	0.614*** (0.110)	0.599*** (0.109)	0.648*** (0.137)	0.691*** (0.120)
Minority X High racial bias state			0.805*** (0.166)				
Minority X Low banking competition				0.293 (0.208)			
Low banking competition				0.052 (0.065)			
Minority X Low dealer competition					0.339** (0.151)		
Low dealer competition					0.094* (0.055)		
Minority X High nonbank financing						0.083 (0.175)	
High nonbank financing						0.197** (0.093)	
Minority X Rural							0.056 (0.223)
Rural							-0.023 (0.078)
Female	-0.397*** (0.052)	-0.259*** (0.039)	-0.265*** (0.038)	-0.260*** (0.039)	-0.259*** (0.039)	-0.259*** (0.039)	-0.259*** (0.039)
Age	-0.014*** (0.003)	0.012*** (0.002)	0.012*** (0.002)	0.012*** (0.002)	0.012*** (0.002)	0.012*** (0.002)	0.012*** (0.002)
log(Income)	-0.228 (0.143)	0.400*** (0.130)	0.396*** (0.130)	0.396*** (0.130)	0.397*** (0.129)	0.397*** (0.130)	0.400*** (0.130)
<i>Auto loan characteristics</i>							
Auto loan term indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes
log(Auto loan amount)	-2.922*** (0.137)	-2.674*** (0.143)	-2.674*** (0.143)	-2.669*** (0.142)	-2.671*** (0.143)	-2.677*** (0.143)	-2.674*** (0.143)
Auto-loan-to-income ratio	-0.326 (0.276)	0.458 (0.281)	0.450 (0.280)	0.443 (0.281)	0.453 (0.281)	0.457 (0.281)	0.458 (0.282)
<i>Credit characteristics</i>							
Credit score _{t-1}		-0.019*** (0.001)	-0.019*** (0.001)	-0.019*** (0.001)	-0.019*** (0.001)	-0.019*** (0.001)	-0.019*** (0.001)
log(Total debt _{t-1})		-0.129*** (0.013)	-0.129*** (0.013)	-0.129*** (0.013)	-0.129*** (0.013)	-0.129*** (0.013)	-0.129*** (0.013)
Debt to income _{t-1}		-0.038* (0.019)	-0.038* (0.019)	-0.037* (0.020)	-0.038* (0.020)	-0.038* (0.020)	-0.038* (0.019)
log(Past due debt _{t-1})		0.336*** (0.015)	0.335*** (0.015)	0.335*** (0.015)	0.336*** (0.015)	0.337*** (0.015)	0.336*** (0.015)
Auto debt share		0.595*** (0.142)	0.594*** (0.142)	0.593*** (0.142)	0.594*** (0.142)	0.597*** (0.143)	0.594*** (0.142)
<i>ZIP code characteristics</i>							
log(Personal income per capita)	0.031 (0.244)	0.071 (0.187)	0.035 (0.182)	0.085 (0.191)	0.065 (0.187)	0.034 (0.175)	0.072 (0.186)
log(Population density)	-0.023 (0.031)	0.010 (0.022)	0.007 (0.021)	0.013 (0.022)	0.010 (0.022)	-0.003 (0.021)	0.006 (0.031)
Bachelor's degree	-2.422*** (0.535)	-0.902*** (0.399)	-0.861*** (0.390)	-0.916*** (0.404)	-0.893*** (0.405)	-0.841*** (0.374)	-0.914*** (0.386)
Commute using car	-1.252*** (0.358)	-0.713*** (0.293)	-0.680*** (0.292)	-0.690*** (0.289)	-0.682*** (0.285)	-0.701*** (0.304)	-0.731*** (0.271)

Table 8
(Continued)

Origination month indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time relative to match indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.255	.440	.441	.441	.441	.441	.440
Observations	25,531	25,523	25,523	25,523	25,523	25,523	25,523

What not to over-read: The interest-rate tests apply only to originated loans. Selection into origination matters: the minority borrowers observed in Table 8 are not the same as all minority applicants. This makes the higher rates even more concerning, but it also means the coefficient is an intensive-margin estimate among approved borrowers.

5.8 Table 9: Borrower race and auto loan default rates

Read:

- In the full sample, the minority default coefficient is negative but not statistically significant.
- Among subprime borrowers, the minority coefficient is about -2.337 percentage points and statistically significant.
- Among prime borrowers, the coefficient is positive but not significant.
- Higher auto loan rates strongly predict default, as expected.

Interpretation: Table 9 is one of the paper's most important tests. If omitted credit risk explains lower approval and higher rates for minorities, minority borrowers should default more after controls. Instead, the subprime minority borrowers who receive loans default less. Since subprime borrowers are closest to the extensive margin of credit provision, this result suggests that marginal minority borrowers may be more profitable than marginal white borrowers.

Economic message: Lower default among subprime minority borrowers is a hallmark of discriminatory rationing: lenders appear to be approving minority borrowers only when they are especially safe, while approving riskier white borrowers at the margin.

What not to over-read: This does not imply every minority borrower is safer than every white borrower. It is a conditional regression statement within the observed approved-loan sample.

Table 9
Borrower race and auto loan default rates

	Auto loan default		
	Full sample (1)	Subprime borrowers (2)	Prime borrowers (3)
<i>Demographics</i>			
Minority	-0.237 (0.397)	-2.337** (1.125)	0.288 (0.345)
Female	0.122 (0.216)	0.619 (1.118)	-0.081 (0.132)
Age	0.016* (0.009)	0.020 (0.049)	0.006 (0.008)
log(Income)	-0.601 (0.450)	-1.734 (1.847)	-0.514 (0.378)
<i>Auto loan characteristics</i>			
Auto loan term indicators	Yes	Yes	Yes
log(Auto loan amount)	1.653*** (0.436)	4.824** (2.104)	0.595* (0.358)
Auto-loan-to-income ratio	-1.697 (1.045)	-3.826 (4.028)	-0.564 (0.902)
Auto loan rate	45.656*** (6.616)	72.553*** (15.369)	16.548*** (5.820)
<i>Credit characteristics</i>			
Credit score _{t-1}	-0.014*** (0.003)	-0.060*** (0.019)	-0.005** (0.002)
log(Total debt _{t-1})	-0.309** (0.145)	-0.707* (0.408)	-0.035 (0.068)
Debt to income _{t-1}	0.261* (0.157)	0.894* (0.494)	0.032 (0.091)
log(Past due debt _{t-1})	0.492*** (0.117)	0.191 (0.168)	0.224* (0.118)
Auto debt share	2.890*** (1.064)	6.579 (4.255)	0.623 (0.604)
<i>ZIP code characteristics</i>			
log(Personal income per capita)	-0.568 (0.802)	-3.414 (3.807)	0.085 (0.511)
log(Population density)	-0.001 (0.099)	0.260 (0.431)	-0.049 (0.054)
Bachelor's degree	0.123 (1.643)	0.086 (8.553)	-0.764 (0.911)
Commute using car	-0.492 (2.176)	13.956 (8.669)	-3.155** (1.539)
Origination month indicators	Yes	Yes	Yes
State-by-year FE	Yes	Yes	Yes
Time relative to match indicators	Yes	Yes	Yes
R-squared	.096	.173	.054
Observations	10,509	2,005	8,480

5.9 Table 10 and Figure 3: The 2013 CFPB intervention

Read:

- Table 10 studies the CFPB's 2013 intervention using outcomes from 2011 to 2017.
- In the auto loan rate columns, the minority rate premium falls after the intervention. The coefficient on Minority $\times$ Post is about -0.490 percentage points in the baseline rate column.
- The triple interaction with high nonbank financing is negative for interest rates, consistent with a stronger effect where indirect auto lending was more relevant.
- The approval columns do not show an offsetting decline in minority access.
- The credit card columns do not show comparable changes, supporting the idea that the auto lending regulatory channel is the relevant one.
- Figure 3 visualizes these yearly patterns: rate disparities decline after the CFPB intervention, especially in high nonbank financing areas, while approval patterns do not show a similar policy-induced collapse.

Interpretation: Table 10 and Figure 3 provide policy-based evidence. The intervention did not randomly assign regulation, but it created a sharp change in oversight of indirect auto lending. The decline in unexplained rate disparities after 2013 suggests that part of the minority rate premium was discretionary and could be reduced by enforcement pressure. The absence of a comparable credit-card response reinforces the auto-specific interpretation.

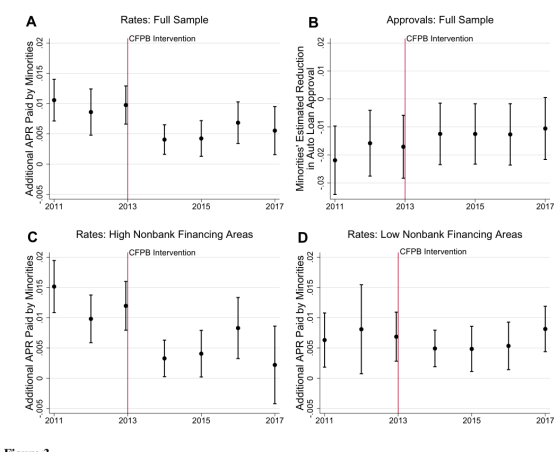
Economic message: Regulatory scrutiny appears to reduce discriminatory pricing without reducing access. This is important because it suggests that the higher minority rates were not necessary to compensate lenders for risk.

What not to over-read: The CFPB intervention evidence is not a clean randomized experiment. Other changes happened over time. The evidence is strongest because the timing, outcome, and high-nonbank-financing heterogeneity all line up with the policy's target.

Table 10
The 2013 CFPB intervention and racial disparities in auto lending

	Auto loan rate			Credit approval (auto)			CC limit increase		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Minority	0.83*** (0.132)	0.614*** (0.205)	0.538*** (0.135)	-1.813*** (0.401)	-2.097*** (0.544)	-1.118** (0.465)	-35.737 (139.563)	29.980 (162.660)	-20.915 (181.790)
Minority X Post	-0.490*** (0.163)	-0.156 (0.233)	-0.403*** (0.175)	0.607 (0.618)	1.451* (0.751)	0.051 (0.750)	173.787 (176.948)	287.217 (210.212)	151.258 (224.936)
Minority X Post X High nonbank financing		-0.625** (0.293)			-1.526 (1.073)			-198.878 (558.361)	
Minority X High nonbank financing		0.401* (0.242)			0.552 (0.811)			-170.850 (313.786)	
Post X High nonbank financing		0.021 (0.150)			-0.739 (0.541)			-218.930 (218.741)	
High nonbank financing		0.139 (0.109)			-0.289 (0.389)			435.434*** (167.489)	
Minority X Post X High racial bias state		-0.312 (0.307)			-1.085 (1.149)			83.548 (337.550)	
Minority X High racial bias state		0.950*** (0.238)			-2.270*** (0.862)			-56.326 (368.967)	
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Auto loan characteristics	Yes	Yes	Yes	n/a	n/a	n/a	n/a	n/a	n/a
ZIP code controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Origination month indicators	Yes	Yes	Yes	n/a	n/a	n/a	n/a	n/a	n/a
State-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time relative to match indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.398	.398	.399	.057	.057	.057	.061	.061	.061
Observations	25,523	25,523	25,523	130,867	130,867	130,867	74,091	74,091	74,091

This table examines the effect of the 2013 CFPB intervention on racial disparities in auto lending using data from 2011 to 2017. Columns 1-3 examine the interest rates on auto loans from our Credit Bureau/HMDA matched panel. The key explanatory variables of interest are *Minority*, and its interactions with indicators for the application occurring in 2014 or later (*Post*), for the person living in a county in the top quartile of nonbank auto lending share (*High nonbank financing*), and for the person living in a state in the top tercile of racial bias based on Google Search Volume for racial slurs (*High racial bias state*). Column 1 presents a difference-in-differences test for whether the CFPB intervention affected the additional interest minorities are charged on auto loans, and columns 2 and 3 present triple-differences tests for whether the CFPB intervention had a larger effect in certain areas (note that several of the interaction terms are subsumed by the state-by-year fixed effects). Columns 4-6 present similar tests examining the effect of the CFPB intervention on auto credit approval. Columns 7-9 conduct a falsification test using credit card limit increases as the dependent variable, following the approach in Table 6. The coefficients are reported in percentage point units for columns 1-6 and dollars for columns 7-9. Standard errors are clustered by state-year. * $p < .1$; ** $p < .05$; *** $p < .01$.



6 Short summary

- The paper studies racial disparities in auto lending by linking credit bureau records to HMDA mortgage data.

- The match allows the authors to observe race, credit characteristics, auto credit outcomes, and subsequent default.
- Minority applicants face lower auto credit access even after rich controls.
- The controlled approval gap is about 1.5 percentage points and is larger for subprime borrowers.
- The approval gap is larger in high racial-bias states and low-competition markets.
- The same borrowers do not show analogous disadvantages in credit card lending.
- Minority borrowers who receive auto loans pay about 70 basis points more after controls.
- Rate disparities are larger in high racial-bias states and low dealer-competition areas.
- Conditional on observables, subprime minority borrowers default less, not more.
- The 2013 CFPB intervention sharply reduced unexplained racial differences in auto loan rates.
- The intervention did not produce an offsetting decline in minority credit access.
- Taken together, the evidence points toward discrimination rather than omitted risk or accurate statistical discrimination.

7 Conclusion

7.1 Core conclusion

Butler, Mayer, and Weston document economically meaningful racial disparities in the U.S. auto loan market. Minority applicants are less likely to obtain auto credit, and minority borrowers who obtain loans pay higher interest rates. These gaps persist after controlling for credit scores, debt, income, geography, time, loan characteristics, and other financial variables.

7.2 Why the paper is persuasive

The strongest feature of the paper is not one regression coefficient. It is the combination of evidence. The approval gap remains after detailed controls. The gap is larger in states with higher racial bias and in less competitive markets. The same borrowers do not face the same pattern in credit card lending. Minority borrowers, especially subprime borrowers, default less after controls. Finally, regulatory scrutiny reduces racial interest-rate gaps. These facts point in the same direction and make a single omitted-variable explanation difficult.

7.3 Economic intuition

The economic intuition is that auto lending contains discretion at several stages: dealer interaction, application processing, indirect financing, markups, and pricing negotiation. When discretion exists, racial bias or stereotypes can affect both whether a borrower obtains credit and the terms of that credit. Competition and regulation can constrain this discretion, which explains why disparities are smaller in competitive markets and after CFPB scrutiny.

7.4 Extensive versus intensive margin

The paper shows discrimination on two margins. On the extensive margin, some minority applicants do not obtain auto loans that observationally similar white applicants obtain. On the intensive margin, minority borrowers who do obtain loans pay higher rates. The combination is important. If only approval differed, one might ask whether pricing offset the approval gap. If only pricing differed, one might ask whether approved borrowers were selected differently. Together with default evidence, the results show that minority borrowers face worse access and worse prices despite not being worse ex post risks.

7.5 Policy implications

The 2013 CFPB evidence suggests that enforcement and oversight can reduce discriminatory pricing. The lack of an offsetting access decline implies that some discriminatory pricing is not required for loan viability. More transparency, stronger monitoring of dealer markups, and clearer reporting of race and loan outcomes could improve market fairness. The paper also suggests that competition policy may matter for discrimination because low-competition markets show larger disparities.

7.6 Caveats and external validity

The paper studies a matched credit bureau/HMDA sample, which is not the entire auto borrowing population. Approval is measured through successful loan origination, so it combines approval and take-up. The paper cannot observe every dealer interaction, vehicle quality, bargaining step, or lender underwriting variable. These caveats are important, but they do not erase the central finding because the evidence is robust across several independent tests.

7.7 Takeaway

The paper’s main contribution is to show that discrimination in consumer credit can be detected even when race is not directly reported in the target market, by carefully linking data sets and using multiple validation tests. Its broader message is that opaque credit markets with human discretion can sustain racial disparities, but competition and regulatory scrutiny can reduce them.